

## A Wireless Hand Gesture-Controlled Wheelchair For Differently Aabled

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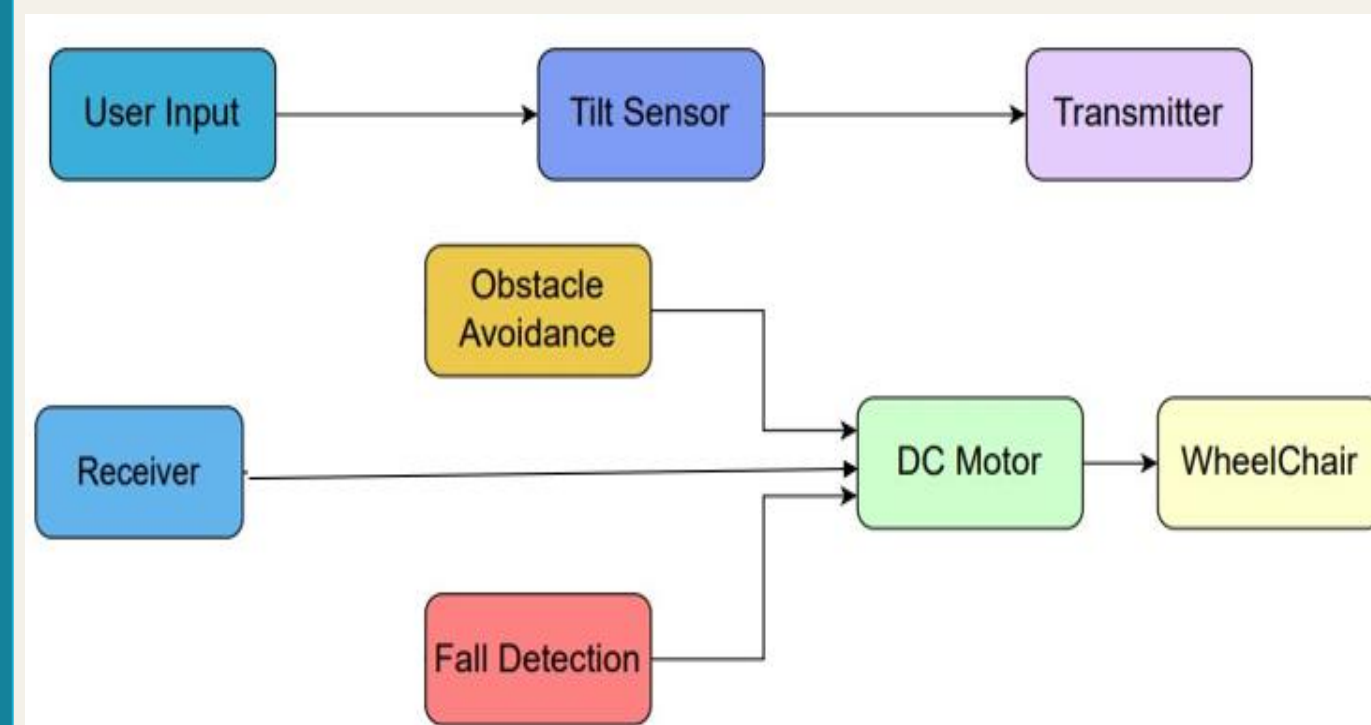
### PROBLEM STATEMENT

Individuals with physical disabilities such as paralysis often face challenges in operating traditional wheelchairs, which require manual effort. For those with limited arm movement or finger dexterity, these methods can be difficult or impractical, increasing their dependence. Current wheelchair control systems lack adaptability, ease of use, and affordability for diverse user needs.

This project proposes a wireless hand gesture-based control mechanism that leverages sensor technology to translate intuitive hand movements into precise mobility commands, enhancing autonomy, comfort, and safety for users.

### APPROACH / DESIGN/ METHODOLOGY

Block Diagram:



#### Methodology:

##### Stage 1: Transmitter – The User

1. **Sensor Input:** 4 tilt sensors are attached to user's hand detect hand gestures by acting as switches that change state based on hand movement.

2. **Transmission of gestures:** Each sensor output, if high triggers a 555 timer oscillator to generate a unique IR frequency for each of the output of the tilt sensors, which modulates and sends data via IR LED.

##### Stage 2: Receiver – The Wheelchair

1. **Reception:** An IR receiver sensor named TSOP48xx, tuned to the specific frequency demodulates the IR signal for each tilt sensor, uses RC filter and PNP transistor switch for clean signal output to the control circuit.

#### Safety Measures:

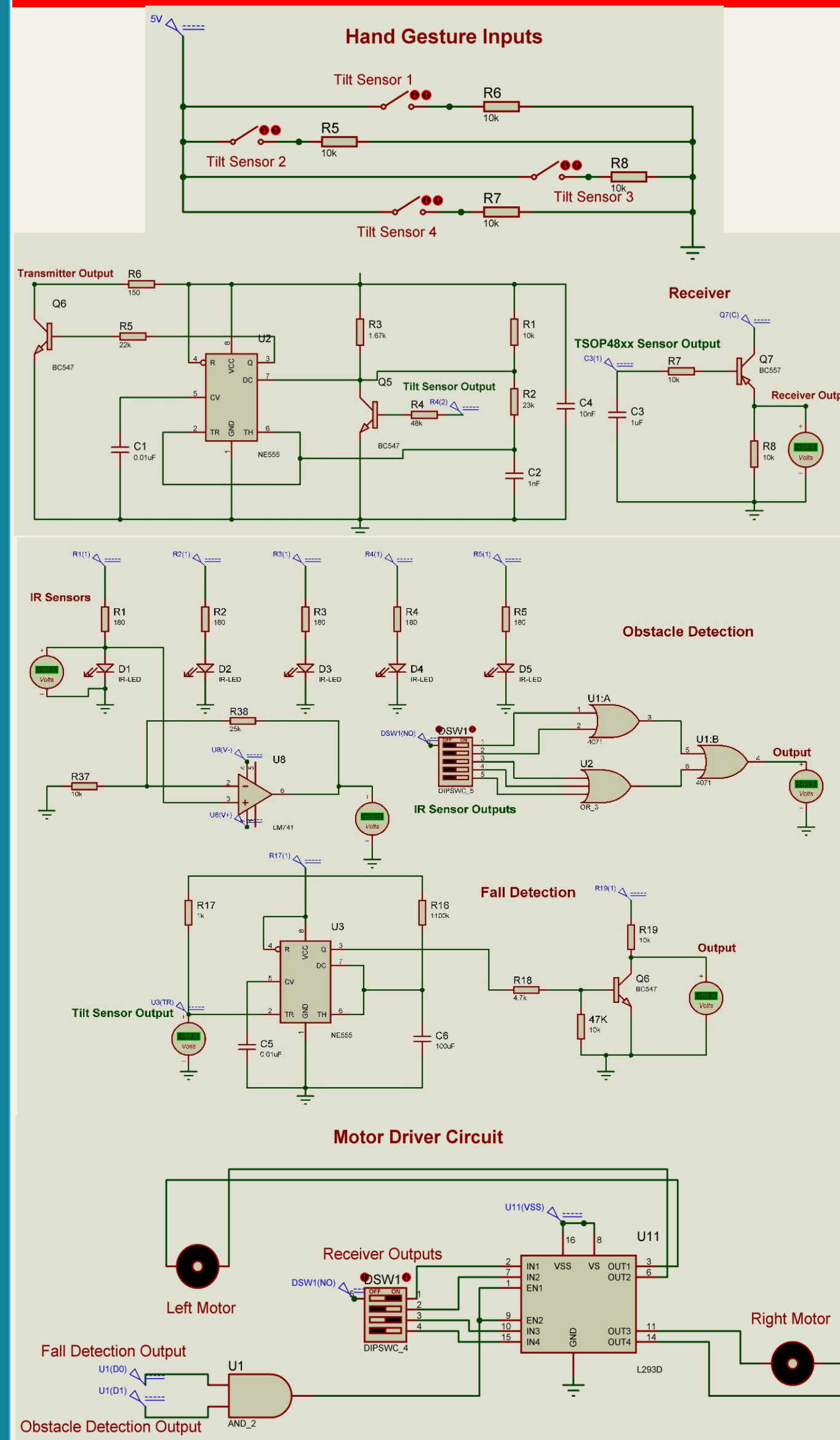
1. **Obstacle Detection:** IR sensors detect obstacles upto knee-length, and logic gates prevent motor operation to avoid collisions.

2. **Fall Detection:** Tilt sensor attached to the wheelchair detect falls, triggering alarms and disabling motors via a 555 timer.

2. **Motor Driver:** L293D IC controls wheelchair motors to move in clockwise/anti-clockwise direction based on the received signals. The obstacle and fall detector circuits control the enable pin of the motor.

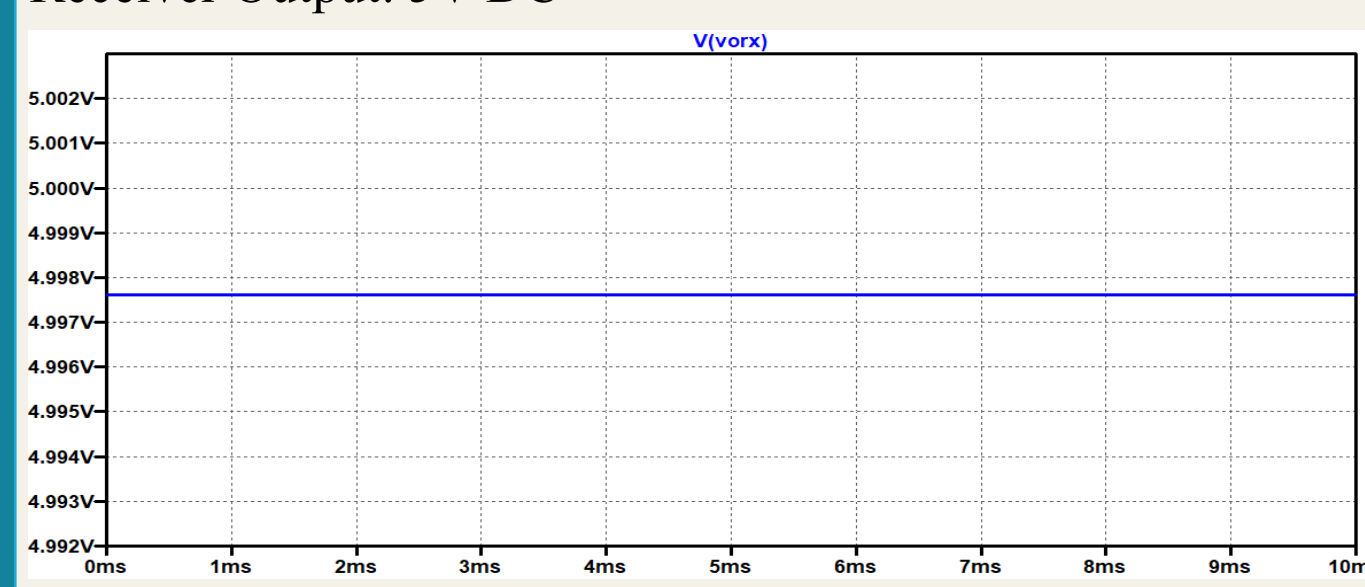
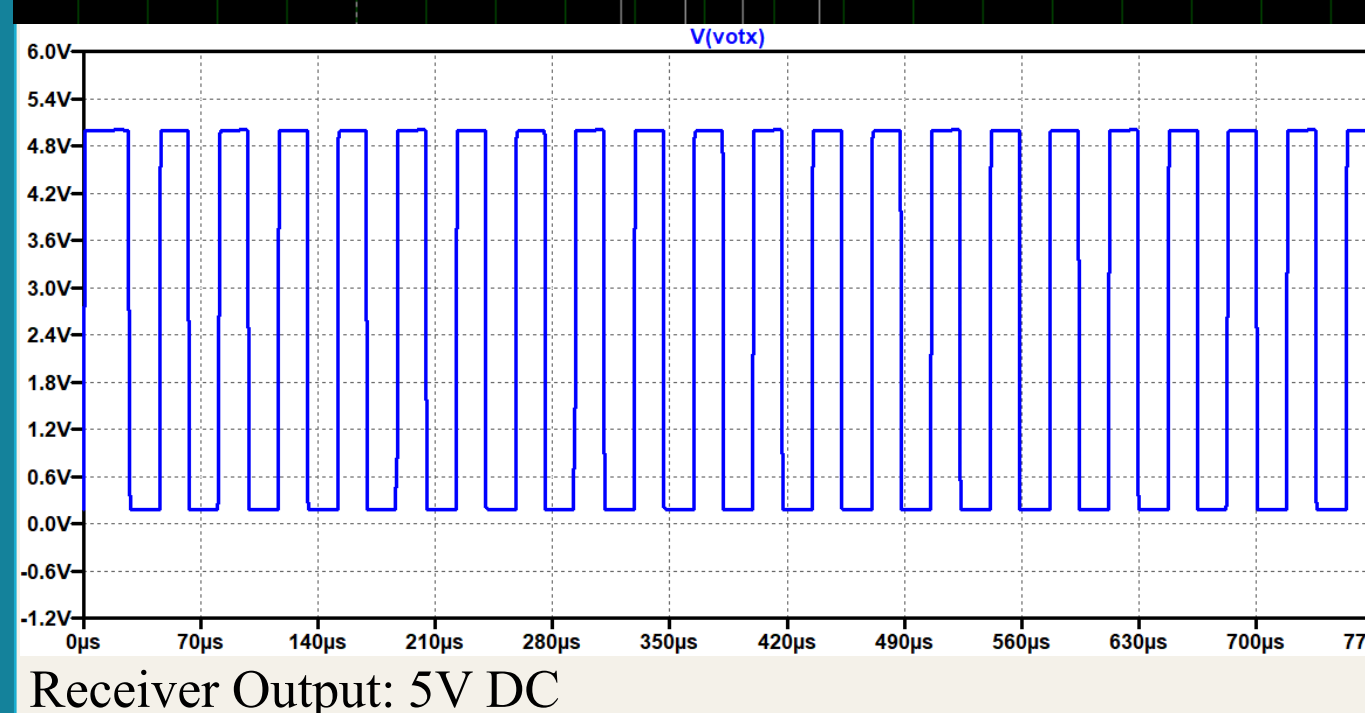
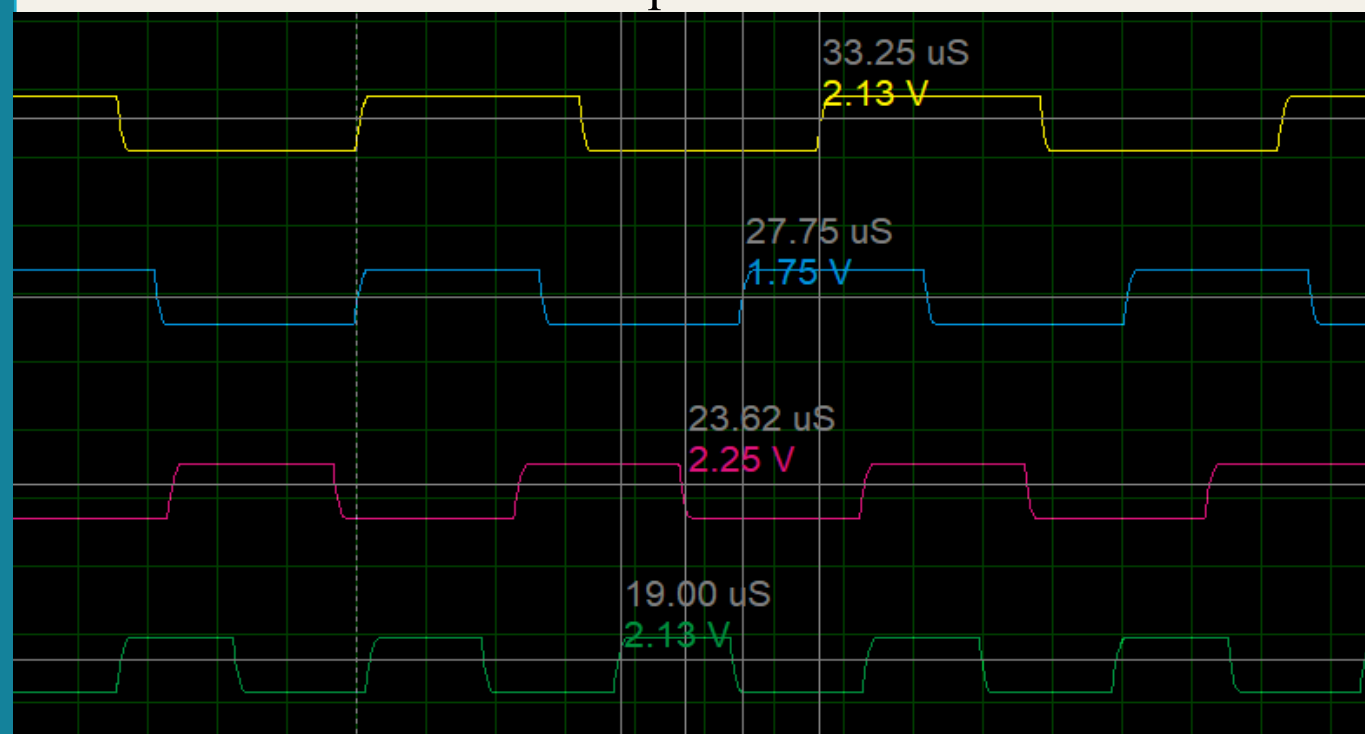
Wireless gesture inputs combined with safety circuits enable efficient and safe wheelchair navigation.

### CIRCUIT DIAGRAM

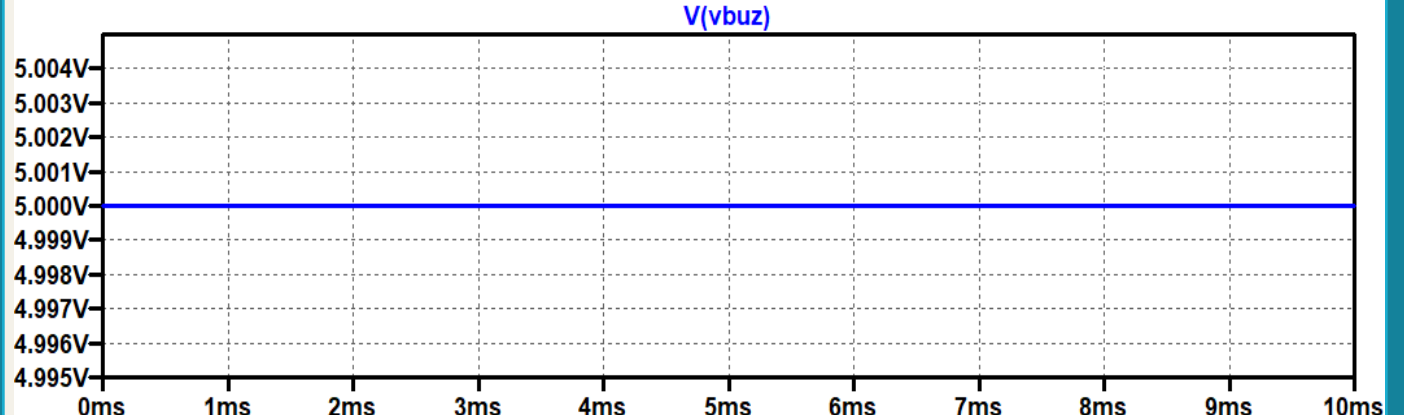


### SIMULATION RESULTS

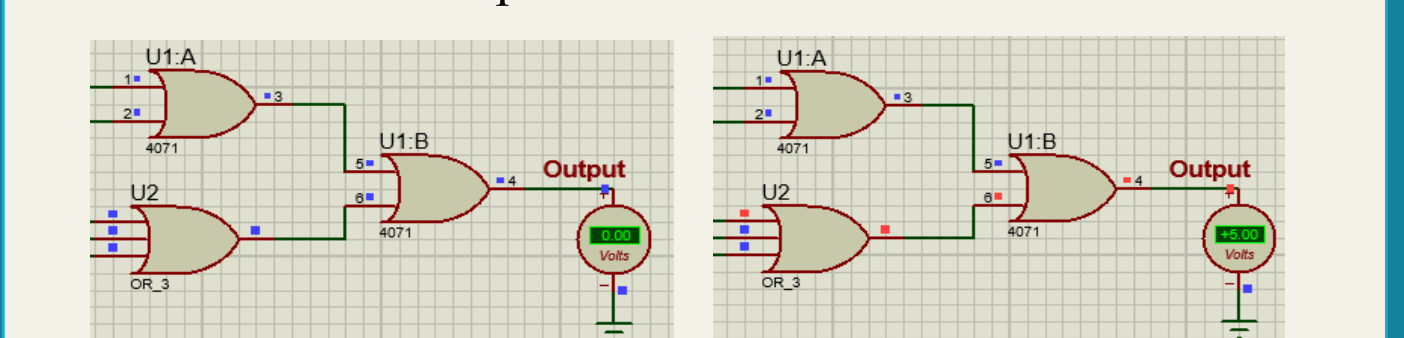
Input from tilt sensors : 5V DC from each  
Transmitter - Modulated Output:



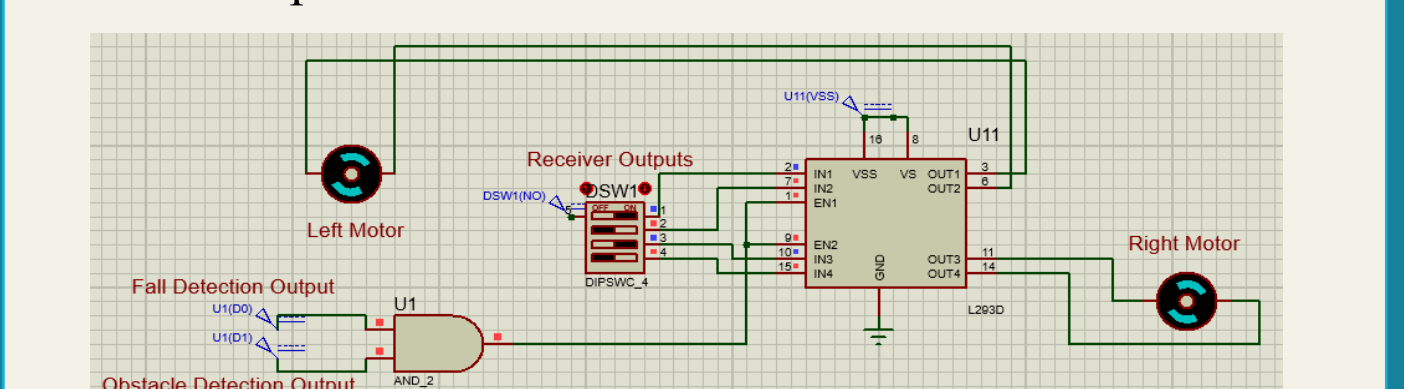
Fall Detector Output:



Obstacle Detector Outputs:



Motor Output:



### RESULTS

- Each tilt sensor(represented by switch) generated a 5V DC signal that was used to trigger the IR transmission circuit.
- Unique IR frequencies, modulated by 555 timer oscillators, were reliably transmitted for each detected gesture, ensuring precise command delivery.
- The 555 timer is set in astable mode to oscillate at the required frequency namely 30k Hz, 36k Hz, 40k Hz and 56k Hz, one to transmit each tilt sensor output.
- The IR receiver (TSOP48xx) successfully demodulated transmitted signals, outputting a clean digital pulse for wheelchair control circuits.
- Obstacle detection using IR sensors produced a digital low output when objects were within knee-height, disabling motor operation for safety.
- Fall detection triggered the alarm circuit and motor cutoff within approximately 100ms, based on tilt sensor activation and rapid response timer circuitry.
- The wheelchair motors, driven by the L293D IC, executed clockwise and anti-clockwise movement based on gesture inputs with reliable logic switching.
- Safety circuits controlled the motor enable pins, preventing operation during obstacle or fall events, ensuring user protection throughout system testing.

### REFERENCES

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